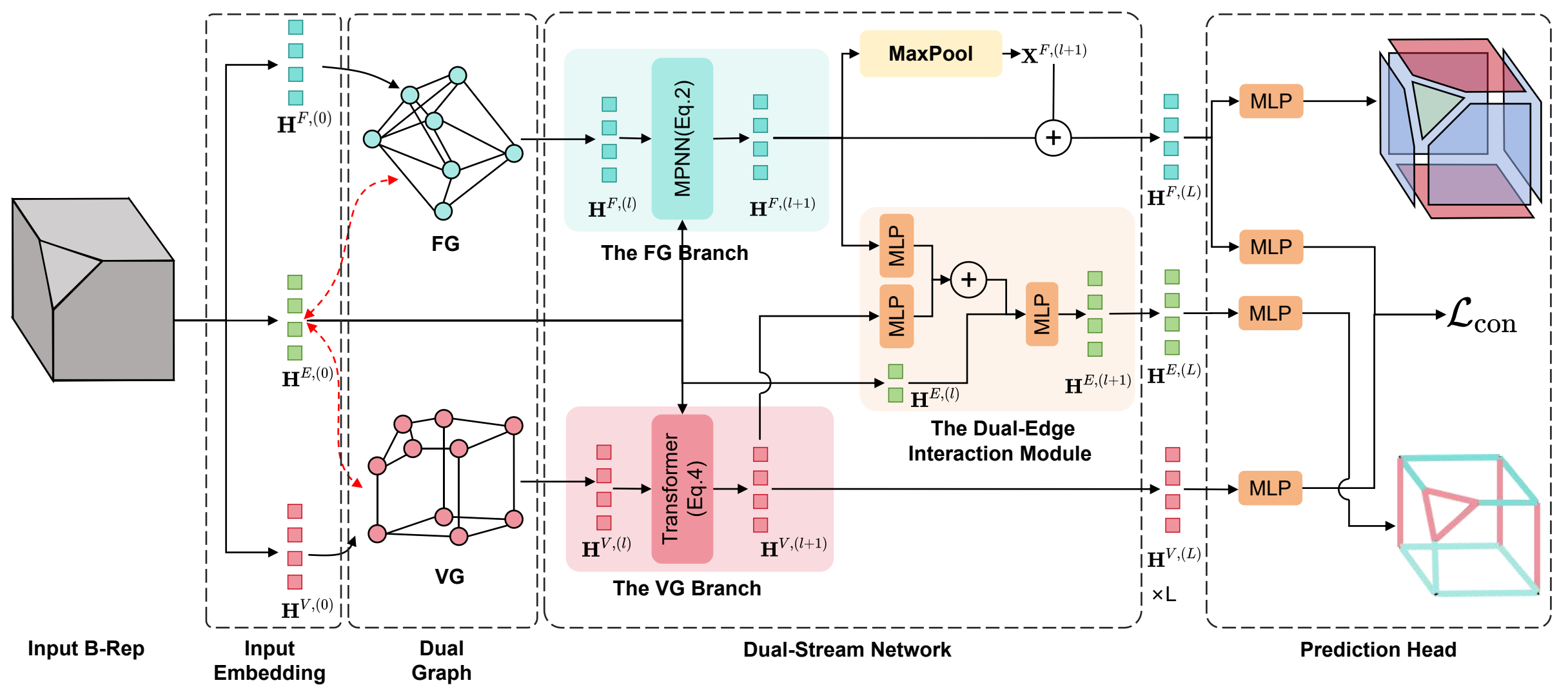

$\mathbf{H}^{F,(0)}$
$\mathbf{H}^{E,(0)}$
$\mathbf{H}^{V,(0)}$
FG
VG
$\mathbf{H}^{F,(l)}$
MPNN(Eq.2)
$\mathbf{H}^{F,(l+1)}$
The FG Branch
MaxPool
$\mathbf{X}^{F,(l+1)}$
MLP
MLP
MLP
$\mathbf{H}^{E,(l)}$
$\mathbf{H}^{E,(l+1)}$
The Dual-Edge Interaction Module
$\mathbf{H}^{V,(l)}$
Transformer (Eq.4)
$\mathbf{H}^{V,(l+1)}$
The VG Branch
$\mathbf{H}^{F,(L)}$
$\mathbf{H}^{E,(L)}$
$\mathbf{H}^{V,(L)}$
×L
MLP
MLP
MLP
$\mathcal{L}_{\text{con}}$
Input B-Rep
Input Embedding
Dual Graph
Dual-Stream Network
Prediction Head